

Finite Excess in Ordinal Nim Multiplication

Exact Order Reductions and the Open 0/1/4 Rule

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Abstract

The finite part of a Kummer carry in Conway's field of ordinal numbers is Lenstra's excess m_p , indexed by an odd prime p . Every presently certified value belongs to $\{0, 1, 4\}$, with the value predicted by the component structure of $\text{ord}_p(2)$. We give an exact reduction of this rule to four universal multiplicative-order assertions. They concern, respectively, a synchronized structural norm, a selected projective class on every ordinary odd Kummer spine, a Gaussian period in a cubic norm tower, and a corrected norm on the exceptional $2 \cdot 3^k$ spine. The reduction retains the full primary valuation in each finite cyclic group; testing only divisibility of an order by p is not sufficient in general.

For each arm we record a concise selected-coordinate normal form. The singleton-even part becomes a Conway–Fermat quotient-order problem and a zero-at-multiples Fibonacci recurrence. The ordinary arm becomes a power-residue symbol of one marked cyclotomic unit. The cubic arm becomes generation by one recursively selected element of a norm-one torus. The exceptional arm becomes irreducibility of an explicit Capelli composition after a corrected quadratic norm. In that exceptional column we also obtain the unconditional level-dependent bound $m_\ell < 16(3^k)^4$ when $f(\ell) = 2 \cdot 3^k$ and $k \geq 2$. We further identify the literal Conway–Fermat points on a fixed supersingular elliptic curve, prove that the ordinary marked primes form a unique inert cyclotomic ray, and show that the cubic recursion and the exceptional auxiliary ancestry are affine forms of one toric Dickson system. We further identify exact proper-conductor packets and marked prime chains in the Conway–Fermat arm, and factor the exceptional selected index into a principal ray kernel and a class-number-controlled circular-unit quotient. These structures sharpen the selected phase which remains to be evaluated; they do not force its nonvanishing. These are reductions, not proofs of the four universal assertions. Formal verification establishes the general group, norm, and polynomial identities and several finite arithmetic certificates; its universal exceptional target is a definition rather than a theorem. Finite computations are reported separately and are not used as evidence for a universal conclusion.

1 Introduction

Conway's nim addition and nim multiplication make the ordinals into a proper class field of characteristic two. Its finite subfields are canonically embedded, and its initial transfinite part is assembled by a sequence of Kummer extensions [4, 8, 9]. Lenstra's description isolates, for each odd prime p , a carry

$$\alpha_p = \kappa_{f(p)} + m_p, \quad f(p) = \text{ord}_p(2),$$

where $\kappa_{f(p)}$ is a structural tower element and the natural number m_p is the *finite excess*. The same construction is part of the general structural description of ordinal fields On_p [6].

Let $\mathcal{Q}(h)$ denote Lenstra's component set for κ_h . The available data suggest the following rigid rule.

Conjecture 1.1 (the 0/1/4 rule). *For every odd prime p ,*

$$m_p = \begin{cases} 0, & \mathcal{Q}(f(p)) \text{ is not a singleton odd prime power,} \\ 4, & f(p) = 2 \cdot 3^k \text{ for some } k \geq 1, \\ 1, & \text{otherwise.} \end{cases}$$

The purpose of this article is to state the exact mathematical frontier of Conjecture 1.1. Theorem 3.1 shows that it is equivalent to four selected order assertions, denoted Z, O, C, D . None of them is known in full. The word *selected* is essential: each field contains many elements with the same degree, trace, norm, or generic cyclotomic shape, while the nim construction chooses one recursively marked Frobenius orbit. The open content is the multiplicative order of that orbit.

The main results are organized as follows. Section 2 gives the exact power and norm criteria. Section 3 states and proves the four-arm equivalence. Sections 4–7 give one lossless normal form for each arm. Section 8 separates the stronger 0/1/4 rule from Lenstra’s boundedness question. Section 9 states the formal and computational boundary.

2 Finite-field and tower preliminaries

For a finite algebraic number x , write

$$d(x) = [\mathbb{F}_2(x) : \mathbb{F}_2].$$

The element κ_h and its component set $\mathcal{Q}(h)$ are those of Lenstra’s tower decomposition. In particular, κ_h is the sum of the selected components κ_q , $q \in \mathcal{Q}(h)$, and $h \mid d(\kappa_h)$. The excess is equivalently

$$m_p = \min \{m \geq 0 : \kappa_{f(p)} + m \text{ is not a } p\text{-th power in its defining finite field}\}.$$

Finite integers in this formula are identified with their canonical finite numbers. We use standard finite-field facts, including cyclicity of the multiplicative group, the norm formula, and Capelli’s irreducibility lemma, as collected in [10].

Proposition 2.1 (exact power criterion). *Let p be a prime, let $p \mid 2^E - 1$, and let $x \in \mathbb{F}_{2^E}^\times$. Then the following are equivalent:*

1. x is a p -th power in $\mathbb{F}_{2^E}^\times$;
2. $x^{(2^E-1)/p} = 1$;
3. $\text{ord}(x) \mid (2^E - 1)/p$.

Consequently, x is not a p -th power if and only if

$$v_p(\text{ord}(x)) = v_p(2^E - 1).$$

Proof. Choose a generator g of the cyclic group of order $2^E - 1$ and write $x = g^a$. The congruence $py \equiv a \pmod{2^E - 1}$ is soluble exactly when $p \mid a$, which is equivalent to $x^{(2^E-1)/p} = 1$. The order formulation follows from $x^n = 1$ if and only if $\text{ord}(x) \mid n$. The final statement retains the full p -primary valuation. $\square$

Remark 2.2 (the valuation cannot be discarded). If $h = f(p)$ and $h \mid E$, lifting the exponent gives

$$v_p(2^E - 1) = v_p(2^h - 1) + v_p(E/h).$$

Thus the shortcut $p \mid \text{ord}(x)$ is equivalent to nonexistence of a p -th root only when the ambient p -part has valuation one. Every statement below uses Proposition 2.1, not the shortcut.

Proposition 2.3 (norm reduction). *Let $h \mid E$, let $p \mid 2^h - 1$, and let $x \in \mathbb{F}_{2^E}^\times$. Then*

$$x^{(2^E-1)/p} = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(x)^{(2^h-1)/p}.$$

In particular, x is a p -th power in $\mathbb{F}_{2^E}$ if and only if its norm is a p -th power in $\mathbb{F}_{2^h}$.

Proof. The norm is $x^{(2^E-1)/(2^h-1)}$. Raising it to $(2^h-1)/p$ gives the displayed identity. Proposition 2.1 gives the last assertion. $\square$

The last equivalence is also visible on Kummer quotients: extension and norm multiply the class by the relative degree. In the applications below, the relevant relative degrees are prime to the current prime, so no selected primary coordinate is lost.

3 The four exact arms

We first define the four order assertions. For $h > 1$, put

$$\Pi_h = \prod_{\substack{\ell \text{ odd prime} \\ \text{ord}_\ell(2)=h}} \ell^{v_\ell(2^h-1)}, \quad T_h = N_{\mathbb{F}_{2^{d(\kappa_h)}}/\mathbb{F}_{2^h}}(\kappa_h).$$

The product Π_h is the complete primary part of $\mathbb{F}_{2^h}^\times$ supported at primes for which h is the first residue degree.

For an odd prime r , let $b_r = d(\alpha_r)$, and for $a \geq 1$ put

$$x_{r,a} = \kappa_{r^a}, \quad F_{r,a} = \mathbb{F}_{2^{r^a b_r}}, \quad y_{r,a} = x_{r,a} + 1,$$

with $F_{r,0} = \mathbb{F}_{2^{b_r}}$. Define the cyclic projective quotient

$$G_{r,a} = F_{r,a}^\times / F_{r,a-1}^\times, \quad L_{r,a} = |G_{r,a}| = \Phi_{r^a}(2^{b_r}).$$

For $k \geq 1$, set $h = 3^k$, let $\zeta_k = \kappa_{3^k}$, and define

$$\gamma_k = \zeta_k + \zeta_k^{-1} \in \mathbb{F}_{2^h}^\times.$$

Finally, put $q = 2^h$, $\omega = \zeta_k^h \in \mathbb{F}_4$, and

$$A_k = \zeta_k^2 + \zeta_k + \omega, \quad M_k = A_k^{q-1}, \quad \Psi_k = \frac{\Phi_{2 \cdot 3^k}(2)}{3}.$$

Theorem 3.1 (four-arm reduction). *Conjecture 1.1 is equivalent to the conjunction of the following four universal assertions.*

Z. For every $h > 1$ such that $\mathcal{Q}(h)$ is not a singleton odd prime power,

$$\Pi_h \mid \text{ord}(T_h).$$

O. For every odd prime $r \neq 3$, every $a \geq 1$, and every prime $p \neq r$ dividing $L_{r,a}$,

$$v_p\left(\text{ord}(y_{r,a}F_{r,a-1}^\times)\right) = v_p(L_{r,a}).$$

C. The element γ_k is primitive in $\mathbb{F}_{2^{3k}}^\times$ for every $k \geq 1$.

D. One has

$$\Psi_k \mid \text{ord}(M_k) \quad (k \geq 1).$$

Proof. Fix an odd prime p and write $h = f(p)$. If $\mathcal{Q}(h)$ is not a singleton odd prime power, the proposed value is zero. Proposition 4.1 below identifies the simultaneous zero claim at residue degree h with Z .

Suppose next that $\mathcal{Q}(h) = \{r^a\}$, where r is odd. Lenstra's degree and component formulas [8, 9] give

$$h = r^a s, \quad s \mid b_r, \quad \kappa_h = x_{r,a}.$$

Proposition 5.1 shows that the row primes are exactly the prime divisors $p \neq r$ of $L_{r,a}$, and that $m_p = 1$ is equivalent to the full primary-order equality in O .

It remains to separate $r = 3$. Here $b_3 = 2$, so $h = 3^a$ or $h = 2 \cdot 3^a$. Proposition 6.1 identifies the first case with primitivity of γ_a , which is *C*. Propositions 7.1 and 7.3 identify the second case with *D*. The cases are disjoint and exhaustive. $\square$

Remark 3.2 (status of the theorem). Theorem 3.1 is an equivalence. It does not establish any of the four universal assertions. The known finite cases and the exact formalization boundary are stated in Section 9.

4 The zero arm

4.1 Structural norm and synchronized phase

Proposition 4.1 (exact support criterion). *Let $h > 1$, let $E = d(\kappa_h)$, and let $T_h = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(\kappa_h)$. For an odd prime p with $f(p) = h$,*

$$m_p = 0 \quad \Longleftrightarrow \quad v_p(\text{ord}(T_h)) = v_p(2^h - 1).$$

Consequently, $m_p = 0$ for every such p if and only if $\Pi_h \mid \text{ord}(T_h)$.

Proof. By definition, $m_p = 0$ exactly when κ_h is not a p -th power in $\mathbb{F}_{2^E}$. Proposition 2.3 moves the test to $T_h \in \mathbb{F}_{2^h}$, and Proposition 2.1 gives the valuation equality. Multiplying the independent primary conditions gives the final assertion. $\square$

When κ_h has several components, the structural norm is not the product of separate component norms. The missing datum is a synchronized Frobenius phase, made explicit by the following general construction.

Proposition 4.2 (synchronized additive resultant). *Let $K = \mathbb{F}_Q$, let $p \mid Q - 1$, and choose $u_i \in L_i = \mathbb{F}_{Q^{r_i}}$ of degree r_i over K . Put*

$$R = \text{lcm}(r_1, \dots, r_t), \quad M = \mathbb{F}_{Q^R}, \quad u = \sum_{i=1}^t u_i,$$

and assume $u \neq 0$ and $p \nmid R$. Let

$$\Omega = \left(\prod_i \mathbb{Z}/r_i\mathbb{Z} \right) / \langle (1, \dots, 1) \rangle.$$

For a phase $\mathbf{d} = (d_i)$, define

$$F_{\mathbf{d}}(Z) = \prod_{j=0}^{R-1} \left(Z + \sum_i u_i^{Q^{j+d_i}} \right).$$

Then $F_{\mathbf{d}} \in K[Z]$, it depends only on the class of $\mathbf{d}$, and

$$\prod_{0 \leq j_i < r_i} \left(Z + \sum_i u_i^{Q^{j_i}} \right) = \prod_{\mathbf{d} \in \Omega} F_{\mathbf{d}}(Z).$$

The selected common embedding is the zero phase, for which

$$F_{\mathbf{0}}(0) = N_{M/K}(u), \quad [F_{\mathbf{0}}(0)] = R[u] \quad \text{in } M^\times/M^{\times p}.$$

Hence u is a non- p -th-power if and only if the selected constant term $F_{\mathbf{0}}(0)$ is a non- p -th-power. Separate component minimal polynomials determine the product over all phases, but not the selected phase when $|\Omega| > 1$.

Proof. Diagonal translation has free orbits of size R on the tuples (j_i) . Frobenius cycles the factors in each orbit, so each orbit product lies in $K[Z]$, and partitioning the tuples proves the factorization. The zero-phase factors are exactly the M/K -conjugates of u , giving the norm. On the p -Kummer quotient the norm exponent is

$$1 + Q + \dots + Q^{R-1} \equiv R \pmod{p}.$$

Since $p \nmid R$, multiplication by R is invertible. Independent Frobenius shifts of the components preserve every separate minimal polynomial while translating the phase, which proves the final limitation. $\square$

In the multicomponent part of Z , the relative degrees supplied by Lenstra's component theorem satisfy the coprimality hypothesis in Proposition 4.2. Thus Z is an order assertion about one distinguished factor, not about the unmarked additive composed product.

There is nevertheless a canonical multiplicative coordinate on the whole primitive-support quotient. It packages all lower-field ambiguity into one cubical cross-ratio without discarding the selected phase.

Proposition 4.3 (cubical primitive-support coordinate). *Let $h > 1$, let S be the set of prime divisors of h , and put $M = \mathbb{F}_{2^h}$. Write $\varphi(x) = x^2$ and $\sigma_r = \varphi^{h/r}$ for $r \in S$. For $A \subseteq S$, put $\sigma_A = \prod_{r \in A} \sigma_r$, and define*

$$H_h = \prod_{r \in S} \mathbb{F}_{2^{h/r}}^\times, \quad \Delta_h(x) = \frac{\prod_{|A| \text{ even}} \sigma_A(x)}{\prod_{|A| \text{ odd}} \sigma_A(x)}.$$

Then

$$\ker \Delta_h = H_h, \quad |\operatorname{im} \Delta_h| = \Phi_h(2),$$

so Δ_h induces an isomorphism

$$M^\times/H_h \simeq \text{im } \Delta_h.$$

In particular, for the selected structural norm T_h , arm Z at degree h is equivalent to

$$\Pi_h \mid \text{ord}(\Delta_h(T_h)).$$

Thus the universal multicomponent problem is the full primary order of one explicit marked cubical cross-ratio.

Proof. The cyclic group $M^\times$ has order $m = 2^h - 1$. The operator above is the power map

$$\Delta_h(x) = x^e, \quad e = \prod_{r|h, r \text{ prime}} (1 - 2^{h/r}).$$

Let $d_r = 2^{h/r} - 1$ and $L = \text{lcm}_{r \in S} d_r$. The subgroup generated by the lower fields has order L , and pairing subsets A and $A \triangle \{r\}$ shows that it lies in $\ker \Delta_h$. It remains to prove $\gcd(m, e) = L$.

Fix an odd prime $\ell \mid m$, put $f = \text{ord}_\ell(2)$, $t = h/f$, and $c = v_\ell(2^f - 1)$. Lifting the exponent gives

$$v_\ell(m) = c + v_\ell(t), \quad v_\ell(d_r) = \begin{cases} 0, & r \nmid t, \\ c + v_\ell(t/r), & r \mid t. \end{cases}$$

If $t = 1$, both the ℓ -adic valuations of $\gcd(m, e)$ and of L are zero. If t is a positive power of ℓ , only $r = \ell$ contributes, and both valuations are $c + v_\ell(t) - 1$. Otherwise some prime $r \neq \ell$ divides t , and both valuations equal the full value $c + v_\ell(t)$. Hence $\gcd(m, e) = L$. The standard cyclotomic product identity now gives $m/L = \Phi_h(2)$, proving the kernel and image claims.

For every prime ℓ occurring in Π_h , one has $\text{ord}_\ell(2) = h$, so the ℓ -primary part of every proper lower-field group is trivial. The quotient map, and therefore Δ_h , preserves the full ℓ -primary order of T_h . Proposition 4.1 gives the last equivalence. $\square$

The raw additive cube behind this proposition does yield a universal nonidentity test. In characteristic two, for edge differences $d_1, \dots, d_s$,

$$\prod_{|A| \text{ even}} \left(b + \sum_{i \in A} d_i \right) + \prod_{|A| \text{ odd}} \left(b + \sum_{i \in A} d_i \right) = \prod_{|A| \text{ odd}} \sum_{i \in A} d_i.$$

Indeed, the even subset sums form a hyperplane W over $\mathbb{F}_2$, and its subspace polynomial $L_W(X) = \prod_{w \in W} (X + w)$ is additive; the left side is $L_W(b) + L_W(b + d_1) = L_W(d_1)$, the odd-coset product. Hence linearly disjoint separated components make the raw cubical coordinate nontrivial. This does not finish Z : T_h is a synchronized norm of the selected sum rather than a separated cube, the norm may kill a nontrivial raw class, and nonidentity is weaker than the full primary order required when Π_h is composite.

4.2 The singleton-even Conway–Fermat problem

Put

$$h_n = 2^{n+1}, \quad c_n = \kappa_{h_n} = 2^{2^n}, \quad E_n = \mathbb{F}_{2^{h_n}}, \quad F_n = 2^{2^n} + 1,$$

and take $E_{-1} = \mathbb{F}_2$. The Conway tower relation is

$$c_n^2 + c_n = a_{n-1}, \quad a_{n-1} = \prod_{j=0}^{n-1} c_j.$$

Let

$$\delta_n = \text{ord}(c_n E_{n-1}^\times) \quad \text{in } E_n^\times / E_{n-1}^\times.$$

Proposition 4.4 (Conway–Fermat form). *One has $1 < \delta_n \mid F_n$. The zero arm at every prime divisor of F_n is equivalent to*

$$\delta_n = F_n.$$

Equivalently, $c_0 c_n$ is primitive in E_n for every n if and only if these equalities hold at every preceding level. This is the selected-order problem studied for Conway towers by Popovych [12].

Proof. The quotient group has order

$$\frac{2^{2^{n+1}} - 1}{2^{2^n} - 1} = F_n,$$

and $c_n \notin E_{n-1}$, so its quotient order is a nontrivial divisor. Every odd prime $p \mid F_n$ has $f(p) = 2^{n+1}$, and its entire primary part in $E_n^\times$ lies in the quotient. Proposition 2.1 therefore gives $m_p = 0$ exactly when the quotient order contains the full p -part. Imposing this at every prime divisor of F_n is equivalent to $\delta_n = F_n$. The primitive-product equivalence follows because the successive quotient orders are supported on pairwise coprime Fermat numbers and c_0 has order three. $\square$

The ordinal choice of the least compatible root is not an additional multiplicative phase. It is exactly a choice of Frobenius gauge.

Proposition 4.5 (all compatible branches form one Frobenius orbit). *Let $\mathcal{T}_n$ be the set of tuples $(x_0, \dots, x_n)$ in $\mathbb{F}_2$ satisfying*

$$x_0^2 + x_0 = 1, \quad x_i^2 + x_i = \prod_{j < i} x_j \quad (1 \leq i \leq n).$$

If $(c_0, \dots, c_n)$ is the literal Conway tuple, then

$$\mathbb{Z}/2^{n+1}\mathbb{Z} \longrightarrow \mathcal{T}_n, \quad j \longmapsto (c_0^{2^j}, \dots, c_n^{2^j})$$

is a bijection. Consequently every compatible binary root-choice string has the same relative order vector $(\delta_0, \dots, \delta_n)$. In particular, changing the top root sends

$$u_n = \frac{c_n + 1}{c_n} \longmapsto u_n^{-1}$$

and does not change δ_n .

Proof. Every prefix has exactly two extensions, because the roots of each displayed quadratic differ by one. Hence $|\mathcal{T}_n| = 2^{n+1}$. Frobenius preserves all the equations, and the displayed tuples are distinct because c_n has exact degree 2^{n+1} . They therefore exhaust $\mathcal{T}_n$. If $q = |E_{i-1}| = 2^{2^i}$, then

$$E_i^\times / E_{i-1}^\times \longrightarrow \ker N_{E_i/E_{i-1}}, \quad [z] \longmapsto z^{q-1}$$

is an isomorphism and sends $[c_i]$ to $c_i^{q-1} = (c_i + 1)/c_i = u_i$. Thus $\text{ord}(u_i) = \delta_i$, and Frobenius preserves the whole relative-order vector. Finally, the nontrivial element over E_{n-1} is $\text{Fr}^{2^n} : x \mapsto x^{2^{2^n}}$; it sends c_n to $c_n + 1$, giving the inverse formula. $\square$

Proper quotient orders admit an exact irreducible-packet description. For odd $e > 1$, let μ_e^{prim} be the primitive e -th roots in $\overline{\mathbb{F}_2}$, put

$$\phi(z) = \frac{z}{(z+1)^2},$$

and define

$$C_e(X) = \prod_{[z] \in \mu_e^{\text{prim}} / (z \sim z^{-1})} (X - \phi(z)) \in \mathbb{F}_2[X].$$

Continue the empty-level convention with $a_{-1} = 1$ and $Q_{-1} = X + 1$.

Proposition 4.6 (primitive-order packet). *The polynomial C_e is monic and squarefree of degree $\varphi(e)/2$. If $e \mid F_n$, then every irreducible factor of C_e has degree 2^n , and, for the degree- 2^n minimal polynomial Q_{n-1} of a_{n-1} ,*

$$\delta_n = e \iff Q_{n-1} \mid C_e.$$

For every odd m , one also has the exact factorization

$$S_m(X) = \prod_{\substack{e \mid m \\ e > 1}} C_e(X).$$

Thus the Fibonacci test is the disjoint union of the exact proper-order packets; it does not mix their phases.

Proof. Cross-multiplication gives

$$\phi(z) = \phi(w) \iff (z+w)(1+zw) = 0,$$

so the fibres are exactly inverse pairs and the definition is unambiguous. Frobenius preserves the root set, proving $C_e \in \mathbb{F}_2[X]$. If $e \mid F_n$, then $2^{2^n} \equiv -1 \pmod{e}$. The order $d = \text{ord}_e(2)$ divides 2^{n+1} but not 2^n , so $d = 2^{n+1}$. The unique involution of $\langle 2 \rangle$ is therefore $-1 = 2^{d/2}$, and the least j with $2^j \equiv \pm 1 \pmod{e}$ is $d/2 = 2^n$. Equivalently every Frobenius orbit on inverse pairs has length 2^n . This proves the factor-degree claim. Since $a_{n-1} = \phi(u_n)$ and $\text{ord}(u_n) = \delta_n$ by Proposition 4.5, the fibre identity gives the selected order equivalence. Finally, the two sides of the last display are monic of degree $(m-1)/2$, and their roots are exactly $\phi(z)$ for $z^m = 1$, $z \neq 1$. $\square$

The packets have a canonical integral lift. Let

$$K_e = \mathbb{Q}(\zeta_e + \zeta_e^{-1}), \quad \vartheta_e = (2 + \zeta_e + \zeta_e^{-1})^{-1}.$$

For odd $e > 1$, the element $1 + \zeta_e$ is a unit because $\Phi_e(-1) = \Phi_{2e}(1) = 1$; hence ϑ_e is an integral unit. Indeed,

$$(1 + \zeta_e)(1 + \zeta_e^{-1}) = 2 + \zeta_e + \zeta_e^{-1}.$$

It generates K_e , since $\zeta_e + \zeta_e^{-1} = \vartheta_e^{-1} - 2$. The minimal polynomial of ϑ_e reduces coefficientwise modulo two to C_e : its conjugates become

$$\frac{1}{z + z^{-1}} = \frac{z}{(z+1)^2}.$$

Since C_e is squarefree, two does not divide $[\mathcal{O}_{K_e} : \mathbb{Z}[\vartheta_e]]$. Dedekind's factorization theorem therefore identifies its irreducible factors precisely with the residue polynomials of ϑ_e at primes above two. Thus a proper value of δ_n is exactly selection of one marked prime factor for a proper conductor $e \mid F_n$; unmarked abelian Frobenius data cannot decide which factor is selected.

The complete downward ancestry of these marked factors also has an exact number-field formulation.

Proposition 4.7 (proper-order prime-chain criterion). *Choose divisors $e_i > 1$ of F_i for $0 \leq i \leq n$, with $e_0 = 3$, and put $L = K_{e_0} \cdots K_{e_n}$. Then*

$$\delta_i = e_i \quad (0 \leq i \leq n)$$

if and only if there is a prime $\mathfrak{P} \mid 2$ of L such that, with $t_i = \vartheta_{e_i} \bmod \mathfrak{P}$,

$$t_0 = 1, \quad t_i^2 + t_{i-1}t_i + t_{i-1}^3 = 0 \quad (1 \leq i \leq n).$$

Hence the all-level Conway–Fermat assertion is precisely the following finite-level statement: for every n , no choice $(e_0, \dots, e_n)$ with at least one $e_i < F_i$ admits such a marked prime chain.

Proof. Assume first that $\delta_i = e_i$. Then u_i is primitive of order e_i and $a_{i-1} = \phi(u_i)$. Since the e_i divide pairwise coprime Fermat numbers, the tuple (u_i) determines, under the Chinese remainder isomorphism of the root groups, one primitive root of order $\prod_i e_i$. Evaluation of the corresponding cyclotomic integer ring in $\overline{\mathbb{F}}_2$ defines a prime above two; restricting it to L gives $\mathfrak{P}$ and $t_i = \vartheta_{e_i} \bmod \mathfrak{P} = a_{i-1}$. The displayed congruence is then the selected resultant identity

$$a_{i-1}^2 + a_{i-2}a_{i-1} + a_{i-2}^3 = 0.$$

This proves necessity.

Conversely, every t_i is nonzero and is a root of C_{e_i} . Each has exact degree 2^i by Proposition 4.6, so their finite residue fields are nested. Put

$$c_{i-1} = t_i/t_{i-1} \quad (1 \leq i \leq n).$$

The congruence gives $c_{i-1}^2 + c_{i-1} = t_{i-1}$, and telescoping gives $\prod_{j < i} c_j = t_i$. Choose c_n as either root of $X^2 + X + t_n$. Then

$$\phi\left(\frac{c_i + 1}{c_i}\right) = t_i \quad (0 \leq i \leq n).$$

Indeed, $\phi((c+1)/c) = c^2 + c$ in characteristic two. Each fibre over a root of C_{e_i} consists of a primitive e_i -th root and its inverse, so the corresponding quotient order is e_i . Proposition 4.5 transfers this compatible branch to the literal Conway branch, proving sufficiency. $\square$

The precise zero-at-multiples coordinate is especially small. Define Fibonacci polynomials over $\mathbb{F}_2$ by

$$S_0(X) = 0, \quad S_1(X) = 1, \quad S_{j+2}(X) = S_{j+1}(X) + XS_j(X).$$

Proposition 4.8 (zero-at-multiples normal form). *For every $n \geq 0$,*

$$\delta_n = \min\{j > 0 : S_j(a_{n-1}) = 0\}.$$

In fact,

$$S_j(a_{n-1}) = 0 \iff \delta_n \mid j.$$

Consequently,

$$\delta_n = F_n \iff S_{F_n/\ell}(a_{n-1}) \neq 0 \text{ for every prime } \ell \mid F_n.$$

Proof. Put $c = c_n$, $a = a_{n-1} = c(c+1)$, and $u = (c+1)/c$. Induction from the recurrence gives the Binet identity

$$S_j(a) = c^j + (c+1)^j = c^j(1 + u^j).$$

The quotient-to-norm-one isomorphism

$$E_n^\times / E_{n-1}^\times \longrightarrow \ker N_{E_n/E_{n-1}}, \quad z E_{n-1}^\times \longmapsto z^{|E_{n-1}^\times|},$$

sends the class of c to u . Thus $\text{ord}(u) = \delta_n$, and the displayed equivalences follow. $\square$

The recurrence is a lossless encoding of the unknown order, not an order bound. Its least zero period is already δ_n . In particular, factorizations, trace identities, or automata derived only from this word reconstruct the selected order rather than force it to be F_n . The classical iterated quadratic tower of Wiedemann has a related norm-one form but a different recursion, so its order results do not transfer by a Frobenius identification [16]. The recent general Artin–Schreier tower theorem of Cagliero, Herman, and Szechtman proves the exact product of the relative orders, but its maximality condition remains precisely that every relative factor δ_j be full; it does not prove $\delta_j = F_j$ [3].

There is nevertheless an exact elliptic realization of this arm. It is useful because it isolates, rather than removes, the remaining Kummer coordinate.

Proposition 4.9 (supersingular realization and pairing boundary). *On*

$$\mathcal{E} : y^2 + y = x^3 + x^2$$

put $P_n = (c_n, c_{n+1})$. Then $P_n \in \mathcal{E}$, and for $n \geq 2$

$$[F_n]P_n = O.$$

More precisely, if π is absolute Frobenius and $\iota = \pi + [1]$, then

$$\iota(x, y) = (x + 1, y + x + 1), \quad \iota^2 = [-1], \quad \pi^4 = [-4].$$

For

$$g = \frac{y + 1}{y}$$

one has

$$g(P_n) = \frac{c_{n+1} + 1}{c_{n+1}}, \quad \text{ord}(g(P_n)) = \delta_{n+1}.$$

If $T = (0, 0)$, then T has order five and

$$\text{div}(g) = 2[-T] + [2T] - 2[T] - [-2T].$$

Thus g is the quotient of the two fixed doubling Miller functions at $-T$ and T , but it is not a Weil or reduced-Tate pairing function attached to the F_n -torsion point P_n .

Proof. The Conway recursion gives

$$c_{n+1}^2 + c_{n+1} = a_n = c_n(c_n^2 + c_n) = c_n^3 + c_n^2,$$

so P_n lies on $\mathcal{E}$. The displayed coordinate formula for $\iota(P) = \pi(P) + P$ follows from the chord law; applying it twice gives the inverse $(x, y) \mapsto (x, y + 1)$. Hence $\pi^2 + 2\pi + 2 = 0$ in $\text{End}(\mathcal{E})$, and squaring this relation gives $\pi^4 = [-4]$.

Put $H = 2^{n+1}$. The 2^H -Frobenius fixes c_n and sends c_{n+1} to $c_{n+1} + 1$, so it sends P_n to $-P_n$. For $n \geq 2$,

$$\pi^H = (\pi^4)^{2^{n-1}} = [2^{2^n}],$$

and therefore $[2^{2^n} + 1]P_n = O$.

The formula for $g(P_n)$ and Proposition 4.8 give its order. The five rational points of $\mathcal{E}/\mathbb{F}_2$ show that T has order five. Direct intersection with the two horizontal lines gives

$$\text{div}(y) = 2[T] + [-2T] - 3[O], \quad \text{div}(y + 1) = 2[-T] + [2T] - 3[O],$$

and subtraction gives the divisor of g . Equivalently, $g = f_{2,-T}/f_{2,T}$ for the $\mathbb{F}_2$ -rational functions normalized at O . Its divisor has coefficients 2, 1, -2, -1 at fixed rational five-torsion, so it cannot be an F_n -Miller function attached to P_n , even modulo an F_n -th power. Any Weil or reduced Tate pairing on $\mathcal{E}[F_n]$ takes values in μ_{F_n} , whereas $g(P_n)$ has order dividing F_{n+1} . The Fermat numbers are pairwise coprime and $\delta_{n+1} > 1$, so these values cannot agree. $\square$

The proposition supplies exact CM annihilator control of the literal Conway point, but generator status still does not follow from annihilation by F_n . For $n \geq 3$, no elliptic endomorphism can carry P_n to P_{n-1} : the order of such an image would divide both coprime numbers F_n and F_{n-1} , although $P_{n-1} \neq O$. For $n = 2$, the same conclusion follows from $\text{ord}(P_1) = 3$ and $\text{ord}(P_2) \mid 17$. The actual ancestry relation $x(P_n) = y(P_{n-1})$ is therefore not an isogeny division tower. The unresolved statement remains exactly the next-level value $\text{ord}(g(P_n)) = F_{n+1}$.

Proposition 4.10 (two-block tensor obstruction). *Let $h = r^a s^b$, where $r \neq s$ are primes and $a, b \geq 1$, and put*

$$M = \mathbb{F}_{2^h}, \quad A = \mathbb{F}_{2^{h/r}}, \quad B = \mathbb{F}_{2^{h/s}}, \quad J = \mathbb{F}_{2^{h/(rs)}}.$$

If $u \in A \setminus J$ and $v \in B \setminus J$, then

$$u + v \notin A^\times B^\times.$$

Moreover,

$$|M^\times / (A^\times B^\times)| = \Phi_h(2).$$

Consequently, if $\Phi_h(2)$ is prime, the class of $u + v$ generates this quotient.

Proof. One has $A \cap B = J$, $[A : J] = s$, and $[B : J] = r$. The relative degrees are coprime, so multiplication identifies $A \otimes_J B$ with M . Choose J -bases of A and B beginning with $(1, u)$ and $(1, v)$. The coefficient matrix of $u \otimes 1 + 1 \otimes v$ has the anti-diagonal minor

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

of nonzero determinant, whereas every element of $A^\times B^\times$ is a pure tensor and has coefficient-matrix rank one. This proves the first assertion. Cyclotomic cancellation gives

$$\frac{(2^h - 1)(2^{h/(rs)} - 1)}{(2^{h/r} - 1)(2^{h/s} - 1)} = \Phi_h(2),$$

and the final assertion follows because a nonidentity element of a group of prime order generates it. $\square$

Proposition 4.11 (five complete multicomponent levels). *Assertion Z holds at*

$$h = 12, 15, 24, 36, 40.$$

Proof. At $h = 12$, the selected element is the sum of components of degrees six and four; take

$$(A, B, J) = (\mathbb{F}_{2^6}, \mathbb{F}_{2^4}, \mathbb{F}_{2^2}).$$

At $h = 24$, the components have degrees six and eight, but the relevant maximal proper support fields are

$$(A, B, J) = (\mathbb{F}_{2^{12}}, \mathbb{F}_{2^8}, \mathbb{F}_{2^4}).$$

The two selected components lie in $A \setminus J$ and $B \setminus J$, respectively. Proposition 4.10 makes the selected class nontrivial, and the prime quotient orders

$$\Phi_{12}(2) = 13, \quad \Phi_{24}(2) = 241$$

make it a generator.

The same proposition applies at $h = 40$. Here the two selected components are $u = \kappa_5 \in \mathbb{F}_{2^{20}} \setminus \mathbb{F}_{2^4}$ and $v = \kappa_8 \in \mathbb{F}_{2^8} \setminus \mathbb{F}_{2^4}$, while $\Phi_{40}(2) = 61681$ is prime. Hence the selected class again generates.

At $h = 36$, put $c = \kappa_3$, $x = \kappa_4$, $y = \kappa_9$, and $z = x + y$. Over $J = \mathbb{F}_{2^6}$, the relations

$$c^6 + c^3 + 1 = 0, \quad x^2 + x = c^3, \quad y^3 = c$$

identify $\mathbb{F}_{2^{36}}$ with $J(y) \otimes_J J(x)$. In the row basis $(1, y, y^2)$ and column basis $(1, x)$, direct reduction gives

$$[z^{37}] = \begin{pmatrix} 1 + c^2 + c^3 & c^2 \\ c^3 + c^4 & c + c^3 + c^4 \\ 0 & c \end{pmatrix}$$

and

$$[z^{109}] = \begin{pmatrix} 1 + c + c^3 & 1 + c + c^2 + c^4 \\ c + c^3 + c^4 + c^5 & c^2 + c^3 + c^4 \\ 1 + c^2 + c^3 + c^5 & c^2 + c^3 + c^4 + c^5 \end{pmatrix}.$$

The upper-left minors are respectively $1 + c^3 + c^4$ and $c^2 + c^3 + c^4 + c^5$, both nonzero because $[\mathbb{F}_2(c) : \mathbb{F}_2] = 6$. Thus neither power lies in $J(y)^\times J(x)^\times$. Since $\Phi_{36}(2) = 4033 = 37 \cdot 109$, the nonidentity of $[z^{37}]$ forces $109 \mid \text{ord}([z])$, while that of $[z^{109}]$ forces $37 \mid \text{ord}([z])$. Thus $[z]$ has full order.

Finally, the level $h = 15$ has a nontrivial synchronized norm. Put $a = \kappa_2$, $c = \kappa_3$, $b = \kappa_4$, and $d = \kappa_5$, so that

$$a^2 + a + 1 = 0, \quad c^3 = a, \quad b^2 + b = a, \quad d^5 = b.$$

The selected element is $\kappa_{15} = c + d$ in $\mathbb{F}_{2^{60}}$. If σ is the 2^{15} -Frobenius, put

$$u = c + \sigma(c) \in \mathbb{F}_{2^3}, \quad v = d + \sigma(d) + \sigma^2(d) + \sigma^3(d) \in \mathbb{F}_{2^5}.$$

Reduction with the four displayed tower relations gives

$$T_{15} = N_{\mathbb{F}_{2^{60}}/\mathbb{F}_{2^{15}}}(c + d) = 1 + v + v^2 + v^3 + u + u^2v + u^2v^3 + u^2v^4.$$

The same reduction gives

$$u = c + c^2 + ac^2, \quad v = d + abd^2 + (1 + ab)d^3 + a(1 + b)d^4.$$

Thus $u^2 + u = ac(c + 1) \neq 0$. Also the coefficient of d^4 in $v^2 + v$ is $a + b \neq 0$; since d has degree five over $\mathbb{F}_{2^4}$, this proves $v^2 + v \neq 0$. Hence u and v have degrees three and five. In the tensor bases $(1, u, u^2)$ and $(1, v, v^2, v^3, v^4)$, the coefficient matrix of T_{15} is

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

Its first two rows and columns have nonzero determinant, so $T_{15} \notin \mathbb{F}_{2^3}^\times \mathbb{F}_{2^5}^\times$. The quotient has prime order $\Phi_{15}(2) = 151$, hence the selected norm generates it. Proposition 4.1 now gives Z at all five levels. $\square$

The universal arm Z therefore remains the simultaneous generation problem for the distinguished synchronized norm, including the all-level Conway–Fermat equalities $\delta_n = F_n$.

5 The ordinary odd Kummer arm

Fix an odd prime r , let $b = b_r = d(\alpha_r)$, and set

$$x_0 = \alpha_r, \quad x_a = \kappa_{r^a} \ (a \geq 1), \quad y_a = x_a + 1, \quad F_a = \mathbb{F}_{2^{r^a b}}.$$

The Kummer equations are $x_1^r = \alpha_r$ and $x_a^r = x_{a-1}$ for $a > 1$.

Proposition 5.1 (projective order criterion). *For every $a \geq 1$,*

$$N_{F_a/F_{a-1}}(y_a) = y_{a-1}.$$

The cyclic quotient $G_{r,a} = F_a^\times / F_{a-1}^\times$ has order

$$L_{r,a} = \frac{2^{r^a b} - 1}{2^{r^{a-1} b} - 1} = \Phi_{r^a}(2^b).$$

If $p \neq r$ is a prime divisor of $L_{r,a}$, then

$$\text{ord}_p(2^b) = r^a, \quad f(p) = r^a s \text{ for some } s \mid b, \quad \mathcal{Q}(f(p)) = \{r^a\}.$$

Conversely, every singleton row with component r^a arises from such a prime. For each corresponding row,

$$m_p = 1 \iff v_p(\text{ord}(y_a F_{a-1}^\times)) = v_p(L_{r,a}).$$

Proof. The relative conjugates of x_a are ξx_a , $\xi \in \mu_r$. Evaluating their binomial at one gives the norm identity. The quotient order follows from the field degrees. For $p \neq r$, the cyclotomic divisor criterion applied to $p \mid \Phi_{r^a}(2^b)$ gives $\text{ord}_p(2^b) = r^a$. Writing $h = f(p)$ then gives $h = r^a s$ with $s \mid b$, and Lenstra's minimality and component theorem give $\mathcal{Q}(h) = \{r^a\}$. The converse is the same argument in reverse.

The prime p does not divide $|F_{a-1}^\times|$, so its full primary contribution lies in the quotient. The selected element x_a has order prime to p , hence is a p -th power; therefore $m_p \geq 1$. Proposition 2.1 applied to y_a proves the final equivalence. $\square$

For a particular row $h = r^a s$, $s \mid b$, the large ambient test may be reduced to the transverse field $K = \mathbb{F}_{2^{r^a s}}$:

$$\Theta_{r,a,s} = N_{F_a/K}(x_a + 1).$$

Then $m_p = 1$ exactly when the order of $\Theta_{r,a,s}$ contains the full p -part of $2^{r^a s} - 1$. By contrast,

$$N_{F_a/F_{r,0}}(x_a + 1) = 1 + \alpha_r$$

lies in a group whose order is prime to p . The lower Kummer norm is therefore blind to the current primary coordinate; the transverse norm is not.

5.1 The marked cyclotomic unit

Let $N_a = \text{ord}(x_a)$, choose a primitive complex N_a -th root ξ_{N_a} , and choose a prime $\mathfrak{P}$ above two whose residue map sends ξ_{N_a} to the selected element x_a . Put $\varepsilon_a = 1 - \xi_{N_a}$. Its reduction is $1 + x_a = y_a$.

Proposition 5.2 (selected residue-symbol form). *Let $p \neq r$ be a prime divisor of $L_{r,a}$. In a cyclotomic field containing μ_{N_a} and μ_p , the following are equivalent:*

1. the projective class of y_a has full p -primary order;
2. y_a is not a p -th power in F_a ;
3. the power-residue symbol satisfies

$$\left(\frac{\varepsilon_a}{\mathfrak{P}}\right)_p \neq 1;$$

4. $\mathfrak{P}$ does not split in the Kummer extension obtained by adjoining a p -th root of ε_a .

Proof. The first two conditions are Proposition 5.1. The usual unramified power-residue formula is

$$\left(\frac{\varepsilon_a}{\mathfrak{P}}\right)_p = \overline{\varepsilon_a}^{(|F_a|-1)/p}.$$

It is nontrivial exactly when the residue is not a p -th power. Kummer theory identifies this symbol with the Frobenius action on a chosen p -th root; see [15]. $\square$

This form explains the real difficulty of O . The conductor and the unmarked conjugate set are determined by the tower, but the conjecture asks for one residue symbol at the prime over two selected by Conway ancestry.

Proposition 5.3 (the selected prime is a unique inert ray). *Put $M = \text{ord}(\alpha_r)$. Choose compatible primitive roots ξ_{N_a} with $\xi_{N_0} \mapsto \alpha_r$ and $\xi_{N_a}^r = \xi_{N_{a-1}}$. Then*

$$N_a = \text{ord}(x_a) = r^a M, \quad \text{ord}_{N_a}(2) = br^a.$$

Let $K_a = \mathbb{Q}(\xi_{N_a})$, and let $\mathfrak{P}_a \mid 2$ be the prime selected by $\xi_{N_a} \mapsto x_a$. For $a \geq 1$, $\mathfrak{P}_a$ is the unique prime of K_a above $\mathfrak{P}_{a-1}$, with relative residue degree r . Moreover,

$$N_{K_a/K_{a-1}}(1 - \xi_{N_a}) = 1 - \xi_{N_{a-1}}.$$

If $p \neq r$ divides $L_{r,a}$, then the lower residue field has trivial p -Kummer quotient. Consequently the residue-field norm

$$F_a^\times / (F_a^\times)^p \longrightarrow F_{a-1}^\times / (F_{a-1}^\times)^p,$$

or equivalently the corresponding finite-field corestriction, sends the selected current class to zero and imposes no restriction on it.

Proof. The irreducible Kummer steps $X^r - x_{a-1}$ increase the r -primary part of the root order by one and preserve every other primary part. This gives $N_a = r^a M$. The defining-field equality $\mathbb{F}_2(x_a) = \mathbb{F}_{2^{br^a}}$ then gives $\text{ord}_{N_a}(2) = br^a$.

The first irreducible binomial also forces $r \mid M$. Hence $[K_a : K_{a-1}] = r$, while the displayed orders show that the residue-degree ratio at the selected primes is also r . Since two is unramified in these odd-conductor cyclotomic fields, the fundamental identity forces one prime above $\mathfrak{P}_{a-1}$, of residue degree r . Taking the product over the r -th roots of unity gives the norm identity.

Finally, $p \mid \Phi_{r^a}(2^b)$ but $p \nmid 2^{br^{a-1}} - 1$. Raising to the p -th power is therefore an automorphism of the lower residue-field multiplicative group, so its p -Kummer quotient is zero. The last assertion follows after reduction at $\mathfrak{P}_{a-1}$; no claim is made that the global cyclotomic-unit class has trivial number-field norm. $\square$

Thus full ancestry leaves no hidden choice of a prime after the birth level: it determines one compatible inert ray. What it does not determine is the new augmentation coordinate of the cyclotomic unit on that ray. Global reciprocity only relates the product of the conjugate symbols, which is one, and ramified conductor formulas at primes above p determine a different local character. A nontrivial ray character may still vanish on the fixed class of $\mathfrak{P}_a$. The missing theorem is therefore a selected-prime Artin-symbol nonvanishing statement, not a decomposition, norm, or conductor calculation.

Proposition 5.4 (phase conservation above the birth level). *Fix a , a prime $p \neq r$ dividing $L_{r,a}$, and for $j \geq a$ put*

$$\Theta_j = (1 + x_j)^{(2^{r^j b} - 1)/p} \in \mu_p.$$

Then

$$\Theta_j = \Theta_a \quad (j \geq a).$$

Thus passage to later levels of the same Kummer spine creates no new current p -coordinate.

Proof. Let

$$S_j = \frac{2^{r^j b} - 1}{2^{r^{j-1} b} - 1}.$$

The norm identity gives $(1 + x_j)^{S_j} = 1 + x_{j-1}$, while

$$\frac{2^{r^j b} - 1}{p} = S_j \frac{2^{r^{j-1} b} - 1}{p}.$$

Therefore $\Theta_j = \Theta_{j-1}$, and induction proves the claim. $\square$

For $r \neq 3$, assertion O is exactly the universal nontriviality of these marked symbols at their birth levels. General lower bounds for orders in finite fields and cyclotomic constructions do not by themselves evaluate the selected residue symbol.

6 The cubic-power arm

Let $h = 3^k$, $\zeta_k = \kappa_{3^k}$, and $\gamma_k = \zeta_k + \zeta_k^{-1}$.

Proposition 6.1 (cyclotomic and half-angle structure). *The element ζ_k is a primitive 3^{k+1} -st root of unity,*

$$\mathbb{F}_2(\zeta_k) = \mathbb{F}_{2^{2h}}, \quad \mathbb{F}_2(\gamma_k) = \mathbb{F}_{2^h}.$$

If s is the inverse of two modulo 3^{k+1} , then

$$\zeta_k + 1 = \zeta_k^s (\zeta_k^s + \zeta_k^{-s}), \quad (\zeta_k^s + \zeta_k^{-s})^2 = \gamma_k,$$

and hence

$$\text{ord}(\zeta_k + 1) = 3^{k+1} \text{ord}(\gamma_k).$$

The translates $\zeta_k + 2$ and $\zeta_k + 3$ have the same order. Consequently, for a prime p with $f(p) = 3^k$, one has

$$m_p = 1 \iff v_p(\text{ord}(\gamma_k)) = v_p(2^h - 1).$$

Assuming γ_{k-1} is primitive, all such rows have excess one exactly when γ_k is primitive.

Proof. The defining Kummer equation gives $\zeta_k^{3^k} = \kappa_2$, a primitive cube root. Since two has order $2 \cdot 3^k$ modulo 3^{k+1} , the cyclotomic polynomial $X^{2h} + X^h + 1$ is irreducible over $\mathbb{F}_2$. The least exponent $j > 0$ with $2^j \equiv \pm 1 \pmod{3^{k+1}}$ is h , which gives the degree of γ_k .

The half-angle identities are direct. Their two factors lie in the coprime circle and real parts of $\mathbb{F}_{2^{2h}}^\times$, of orders dividing $2^h + 1$ and $2^h - 1$, respectively. Squaring preserves odd order, so the order formula follows. The other two translates are Frobenius conjugate to the same decomposition. Finally, a prime with $f(p) = h$ cannot divide the 3-power factor, so Proposition 2.1 gives the stated criterion. $\square$

The selected Gaussian period obeys

$$\gamma_k^3 + \gamma_k = \gamma_{k-1}, \quad N_{\mathbb{F}_{2^{3^k}}/\mathbb{F}_{2^{3^{k-1}}}}(\gamma_k) = \gamma_{k-1}.$$

This propagates every old primary factor. It does not control the new factor $\Phi_{3^k}(2)$.

Proposition 6.2 (selected norm-one generator). *Put*

$$q_k = 2^{3^{k-1}}, \quad U_k = \ker\left(N_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}}\right), \quad \eta_k = \gamma_k^{q_k-1},$$

and set $\eta_0 = 0$. Then

$$|U_k| = q_k^2 + q_k + 1 = \Phi_{3^k}(2), \quad \mathbb{F}_2(\eta_k) = \mathbb{F}_{2^{3^k}},$$

and η_k is the selected root of the irreducible cubic

$$X^3 + \eta_{k-1}X^2 + (\eta_{k-1} + 1)X + 1.$$

Equivalently,

$$N(\eta_k) = 1, \quad \text{Tr}(\eta_k) = \eta_{k-1}, \quad e_2(\eta_k) = \eta_{k-1} + 1.$$

Moreover,

$$C_k \iff C_{k-1} \text{ and } \text{ord}(\eta_k) = q_k^2 + q_k + 1,$$

where C_j denotes primitivity of γ_j .

Proof. The element η_k has norm one by its definition. A direct Frobenius calculation from $\zeta_k^{q_k}$ gives its displayed cubic; its symmetric coefficients give the norm, trace, and second elementary symmetric function. It cannot lie in the base field, since a base-field element of U_k has order dividing $\gcd(q_k - 1, 3) = 1$. Hence the cubic is irreducible. Since its quadratic coefficient recovers η_{k-1} , induction from $\eta_0 = 0$ gives $\mathbb{F}_2(\eta_k) = \mathbb{F}_{2^{3^k}}$.

The cyclic group $\mathbb{F}_{q_k^3}^\times$ splits as the product of $\mathbb{F}_{q_k}^\times$ and U_k , whose orders are coprime because $q_k \equiv 2 \pmod{3}$. The norm determines the old factor of γ_k , while exponentiation by $q_k - 1$ detects the new U_k -factor without changing its order. This proves the last equivalence. $\square$

The cubic ancestry admits a closed rational-dynamical form. This makes the word “selected” in Proposition 6.2 completely explicit.

Proposition 6.3 (Singer trace ancestry and Möbius conjugacy). *Put $q = 2^{3^{k-1}}$, $N = q^2 + q + 1$, and*

$$\mathcal{D}_k = \{x \in U_k : x^{q+1} + x + 1 = 0\}.$$

For $x \in \mathcal{D}_k$, define

$$R(x) = \text{Tr}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(x) = \frac{x^3 + x + 1}{x^2 + x}.$$

If $\mathcal{A}_j = \{\eta_j^{2^i} : 0 \leq i < 3^j\}$ and $\mathcal{A}_0 = \{0\}$, then

$$\mathcal{D}_k \cap R^{-1}(\mathcal{A}_{k-1}) = \mathcal{A}_k.$$

Thus the complete lower ancestry selects one unique absolute-Frobenius orbit inside the Singer set $\mathcal{D}_k$.

If Q_j is the minimal polynomial of η_j , then, with $A = X^2 + X$ and $B = X^3 + X + 1$,

$$Q_k(X) = A^{3^{k-1}} Q_{k-1}(B/A), \quad Q_0(X) = X.$$

Over $\mathbb{F}_4$, choose $\omega^2 + \omega + 1 = 0$ and put

$$W(x) = \frac{x + \omega}{x + \omega^2}.$$

Then

$$W(R(x)) = W(x)^3.$$

Consequently $z_k = W(\eta_k)$ has exact order 3^{k+1} , and in fact the orientation fixed by the Conway root is

$$z_k = \zeta_k^2.$$

In particular,

$$\eta_k = \omega^2 \frac{z_k + \omega^2}{z_k + 1}.$$

The current order assertion is equivalent to the family of literal polynomial nondivisibilities

$$\text{ord}(\eta_k) = N \iff Q_k(X) \nmid X^{N/\ell} - 1 \text{ for every prime } \ell \mid N.$$

Proof. For $x \in \mathcal{D}_k$, its defining equation gives

$$x^q = \frac{x + 1}{x}, \quad x^{q^2} = \frac{1}{x + 1}.$$

Adding the three conjugates gives the displayed formula for R . The fibre over t is exactly the root set of

$$X^3 + tX^2 + (t + 1)X + 1.$$

Proposition 6.2 therefore shows that the fibre over each $\eta_{k-1}^{2^i}$ consists of the three conjugates $\eta_k^{2^{i+a}3^{k-1}}$, $0 \leq a < 3$. These fibres exhaust $\mathcal{A}_k$, proving the orbit identity. Taking their product gives the denominator-cleared recursion for Q_k .

Direct expansion using $\omega^2 + \omega + 1 = 0$ gives

$$\frac{R(x) + \omega}{R(x) + \omega^2} = \frac{(x + \omega)^3}{(x + \omega^2)^3}.$$

There are no poles along the selected iteration: $\eta_j \notin \mathbb{F}_2$ for $j \geq 1$, and $\mathbb{F}_{2^{3j}} \cap \mathbb{F}_4 = \mathbb{F}_2$, so neither denominator $\eta_j^2 + \eta_j$ nor $\eta_j + \omega^2$ vanishes. Since $R^{\circ k}(\eta_k) = 0$ and $W(0) = \omega^2$, one has $z_k^{3^k} = \omega^2$,

hence z_k has exact order 3^{k+1} . The selected orientation can be read without ambiguity. If $h = 3^k$, then

$$q = 2^{3^{k-1}} \equiv h - 1 \pmod{3h},$$

where the congruence follows from the case $2 \equiv 2 \pmod{9}$ by successive cubing. Therefore $\zeta_k^q = \omega \zeta_k^{-1}$. With $y = \zeta_k^2$, direct substitution gives

$$\eta_k = \frac{\gamma_k^q}{\gamma_k} = \frac{\omega + \omega^2 y}{1 + y} = \omega^2 \frac{y + \omega^2}{y + 1}.$$

Consequently $\eta_k + \omega = y/(1 + y)$ and $\eta_k + \omega^2 = 1/(1 + y)$, whence $W(\eta_k) = y = \zeta_k^2$. Finally, irreducibility of Q_k , followed by the prime-divisor test in the cyclic group of order N , gives the last equivalence. $\square$

This proposition identifies the literal selected orbit rather than an ambient degree or norm class. It does not prove primitivity: the conjugacy turns the remaining assertion into the power-residue order of one marked Möbius cyclotomic unit. In particular, the lower trace ancestry is nonlinear, while the relative norm of every element of $\mathcal{D}_k$ is one; no multiplicative descent to the preceding level survives.

The obstruction can also be separated exactly into a circular-unit term and a principal ray term. This is useful because it identifies which part is not controlled by ordinary class-number information.

Proposition 6.4 (exact relative circular-ray obstruction). *For $k \geq 1$, put*

$$L = \mathbb{Q}(\zeta_{3^{k+1}})^+, \quad K = \mathbb{Q}(\zeta_{3^k})^+, \quad h = 3^k, \quad q = 2^{h/3}.$$

For $F = K, L$, let $\mathfrak{p}_F = 2\mathcal{O}_F$, let $k_F = (\mathcal{O}_F/\mathfrak{p}_F)^\times$, and let E_F and C_F be respectively the global units and the circular units, with -1 included in C_F . Write bars for reduction modulo $\mathfrak{p}_F$, and set

$$U = \ker(N : k_L \rightarrow k_K), \quad T_F = \overline{E}_F/\overline{C}_F, \quad S_F = k_F/\overline{C}_F, \quad R_F = k_F/\overline{E}_F.$$

Then $|U| = q^2 + q + 1$, the selected element satisfies $\eta_k = \overline{\tau(\gamma_k)}/\gamma_k$ for $\tau(\zeta) = \zeta^{2^{h/3}}$, and there is a canonical isomorphism

$$U/\langle \eta_k \rangle \simeq \ker(N : S_L \rightarrow S_K).$$

All norm maps in the rows

$$1 \longrightarrow T_F \longrightarrow S_F \longrightarrow R_F \longrightarrow 1$$

are surjective, and hence there is an exact sequence

$$1 \longrightarrow \ker(N : T_L \rightarrow T_K) \longrightarrow U/\langle \eta_k \rangle \longrightarrow \ker(N : R_L \rightarrow R_K) \longrightarrow 1.$$

In particular,

$$[U : \langle \eta_k \rangle] = |\ker(N : R_L \rightarrow R_K)| \frac{[\overline{E}_L : \overline{C}_L]}{[\overline{E}_K : \overline{C}_K]}, \quad (*)$$

where R_F is canonically the principal kernel

$$\ker(\text{Cl}_F(\mathfrak{p}_F) \longrightarrow \text{Cl}_F).$$

Moreover, the second factor in $()$ divides h_L^+/h_K^+ , the ratio of the ordinary real class numbers. Consequently arm C is equivalent to simultaneous triviality of the relative principal 2-ray kernel and the relative reduced unit/circular-unit kernel.*

Proof. Let $\sigma(\zeta) = \zeta^2$, put $\tau = \sigma^{h/3}$, and use the standard circular units

$$c_a = \zeta^{(1-a)/2} \frac{1 - \zeta^a}{1 - \zeta}.$$

Here the half-exponent is interpreted modulo 3^{k+1} . The circular units are generated by signs and the c_a with $3 \nmid a$, while the powers of 2 traverse these classes modulo sign [15]. The relations

$$\sigma^j(c_2) = \frac{c_{2^{j+1}}}{c_{2^j}}, \quad \sigma(c_2) = \frac{c_4}{c_2} = \gamma_k$$

show that the conjugates of γ_k generate C_L modulo signs. The circular-unit distribution relation gives $N_{L/K}(\gamma_k) = \gamma_{k-1}$, so the circular-unit norm is onto. The reductions of $(\tau - 1)C_L$ generate $\langle \eta_k \rangle$. If $D_C = \ker(N : C_L \rightarrow C_K)$, then, in additive module notation,

$$3d = (1 - \tau)(2 + \tau)d \quad (d \in D_C).$$

Thus $D_C/(\tau - 1)C_L$ is killed by 3. Since $3 \nmid |U|$, reduction gives $\overline{D_C} = \langle \eta_k \rangle$. The same cube-adjustment $c \mapsto c^3/N(c)$ shows that this is the full kernel of the norm on $\overline{C_L}$. The snake lemma applied to $1 \rightarrow \overline{C_F} \rightarrow k_F \rightarrow S_F \rightarrow 1$ now gives the first isomorphism.

The finite-field norm is onto. Norm on $\overline{E_L}$ is also onto: an embedded base unit has relative norm its cube, and cubing is an automorphism because $3 \nmid |k_K|$. For these real 3-power cyclotomic fields, Sinnott's index formula and Iwasawa's 3-indivisibility give

$$|E_F/C_F| = h_F^+, \quad 3 \nmid h_F^+;$$

see [15]. Hence the same embedded-cube argument makes the norm on $T_L \rightarrow T_K$ onto. The remaining surjectivities and the short exact sequence follow from the snake lemma, and taking orders gives (*).

Finally, the global circular norm is onto and

$$D_E/D_C \simeq \ker(N : E_L/C_L \rightarrow E_K/C_K),$$

whose order is h_L^+/h_K^+ . Reduction onto the corresponding kernel in $T_L \rightarrow T_K$ follows by first correcting the residue norm with a circular unit and then applying $e \mapsto e^3/N(e)$. Therefore the reduced kernel has order dividing the class-number ratio. The ray-class interpretation of R_F is the usual ray exact sequence. $\square$

Even $h_L^+ = h_K^+ = 1$ would kill only the T -factor in (*); the principal 2-ray factor remains exactly the marked residue-order problem. Conversely, arm C does not imply class-number one. Thus ordinary circular-unit index theorems sharpen the location of the obstruction but do not supply its missing selected ray coordinate.

Proposition 6.5 (exact Capelli dichotomy). *Let $Q_k \in \mathbb{F}_2[X]$ be the minimal polynomial of η_k , and let ℓ be a prime divisor of $N_k = \Phi_{3^k}(2)$. Exactly one of the following occurs:*

1. $\eta_k \notin (\mathbb{F}_{2^{3^k}}^\times)^\ell$, and $Q_k(X^\ell)$ is irreducible of degree $3^k\ell$;
2. $\eta_k \in (\mathbb{F}_{2^{3^k}}^\times)^\ell$, and $Q_k(X^\ell)$ is the product of exactly ℓ distinct irreducible polynomials, each of degree 3^k .

In particular, both the successful and failed cases have an odd number of irreducible factors. Factor-count parity, and hence any criterion which sees only that parity, cannot decide the current C -coordinate.

Proof. Put $h = 3^k$. The preceding proposition gives $\mathbb{F}_2(\eta_k) = \mathbb{F}_{2^h}$, while $\ell \mid 2^h - 1$ gives $\mu_\ell \subset \mathbb{F}_{2^h}$. Capelli's lemma and the prime-degree Kummer criterion therefore say that $Q_k(X^\ell)$ is irreducible exactly when η_k is not an ℓ -th power.

Suppose instead that $b^\ell = \eta_k$ in $\mathbb{F}_{2^h}$. Every root of $Q_k(X^\ell)$ is an ℓ -th root of a Frobenius conjugate of η_k , hence belongs to $\mathbb{F}_{2^h}$, because that field also contains μ_ℓ . Such a root has degree at least h over $\mathbb{F}_2$, since its ℓ -th power has degree h , and therefore has degree exactly h . The composition is separable and has degree $h\ell$, so it has exactly ℓ distinct irreducible factors, all of degree h . Finally ℓ is odd because it divides $2^h - 1$. $\square$

For a prime $\ell \mid |U_k|$, put

$$\Omega_{k,\ell} = \eta_k^{|U_k|/\ell}.$$

The exact terminal test is

$$\text{ord}(\eta_k) = |U_k| \iff \Omega_{k,\ell} \neq 1 \text{ for every prime } \ell \mid |U_k|.$$

This formulation again retains the full ℓ -primary valuation. The minimal polynomial of γ_k is also explicit. If $D_n(X, 1)$ is the Dickson polynomial defined by $D_n(t + t^{-1}, 1) = t^n + t^{-n}$, then

$$P_k(X) = D_{3^k}(X, 1) + 1 = (X^3 + X)^{o_k} + 1$$

is irreducible of degree 3^k , and C_k says precisely that it is a primitive polynomial. Irreducibility is therefore already built into the tower; primitivity is the open statement. General lower bounds for multiplicative orders of Gaussian periods, such as [1], do not give maximal order for this recursively selected period.

In fact degree is maximally insensitive here. Every prime $\ell \mid \Phi_{3^k}(2)$ satisfies $\text{ord}_\ell(2) = 3^k$. If $z \neq 1$ lies in U_k , choose such a prime $\ell \mid \text{ord}(z)$. Then $[\mathbb{F}_2(z) : \mathbb{F}_2]$ is both divisible by $\text{ord}_\ell(2) = 3^k$ and at most 3^k , hence equals 3^k . Thus even an element of proper order in U_k has the same full field degree as a generator.

The recursion in Proposition 6.2 is lossless. It reconstructs every finite selected truncation up to Frobenius, and Frobenius preserves multiplicative order. Trace, norm, and the sparse cubic therefore identify the correct selected coordinate but do not evaluate $\Omega_{k,\ell}$. This is the exact remaining content of arm C .

7 The exceptional arm

Let $h = 3^k$, $q = 2^h$, $\zeta = \zeta_k$, and $\omega = \zeta^h \in \mathbb{F}_4$.

Proposition 7.1 (unconditional lower bound). *If p is a prime with $f(p) = 2 \cdot 3^k$, then $m_p \geq 4$.*

Proof. Such a prime divides $2^h + 1$ and divides neither 3^{k+1} nor $2^h - 1$. The element ζ has order 3^{k+1} , while Proposition 6.1 gives

$$\text{ord}(\zeta + m) = 3^{k+1} \text{ord}(\gamma_k) \quad (m = 1, 2, 3),$$

with $\text{ord}(\gamma_k) \mid 2^h - 1$. Hence all four translates $m = 0, 1, 2, 3$ have order prime to p , and each is a p -th power in its defining field. The definition of the excess gives the claim. $\square$

The translate by four lies in the compositum with $\mathbb{F}_{16}$, so its first useful invariant is a quadratic norm rather than an order calculation in $\mathbb{F}_{2^{2h}}$.

Proposition 7.2 (corrected norm). *In $\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}$, the nontrivial relative Frobenius sends the finite number 4 to 5, and*

$$N_{\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}}(\zeta + 4) = (\zeta + 4)(\zeta + 5) = \zeta^2 + \zeta + \omega = A_k.$$

Proof. The intersection of $\mathbb{F}_{16}$ and $\mathbb{F}_{2^{2h}}$ is $\mathbb{F}_4$, so the relative Frobenius is the nontrivial automorphism of $\mathbb{F}_{16}/\mathbb{F}_4$ and exchanges 4 and 5. These are the two roots of $Y^2 + Y + \omega$; hence $4 \cdot 5 = 4^2 + 4 = \omega$. Expanding the product proves the formula. $\square$

Define

$$R_k(Y) = \text{Res}_X \left(X^{2h} + X^h + 1, Y + X^h + X^2 + X \right) \in \mathbb{F}_2[Y].$$

Theorem 7.3 (exact exceptional target). *The polynomial R_k is the irreducible degree- $2h$ minimal polynomial of A_k . The prime divisors of*

$$\Psi_k = \frac{\Phi_{2 \cdot 3^k}(2)}{3}$$

are exactly the primes $\ell \neq 3$ with $\text{ord}_\ell(2) = 2h$, with their full valuations in $2^{2h} - 1$. The following are equivalent:

1. $m_\ell = 4$ for every prime ℓ with $f(\ell) = 2 \cdot 3^k$;
2. $\Psi_k \mid \text{ord}(M_k)$, where $M_k = A_k^{q-1}$;
3. A_k is not an ℓ -th power in $\mathbb{F}_{2^{2h}}$ for every prime $\ell \mid \Psi_k$;
4. $R_k(X^\ell)$ is irreducible over $\mathbb{F}_2$ for every prime $\ell \mid \Psi_k$.

Proof. The element ζ has minimal polynomial $X^{2h} + X^h + 1$. Suppose $A_k^{2^j} = A_k$, and reduce $a = 2^j$ modulo $3h$. The resulting relation on ζ has support the symmetric difference of

$$\{ah, a, 2a\} \quad \text{and} \quad \{h, 1, 2\}.$$

Every multiple of $X^{2h} + X^h + 1$ of degree below $3h$ has support a disjoint union of triples $\{r, r + h, r + 2h\}$. If $a \equiv 2 \pmod{3}$, the support contains $h, 2h$ but not zero. If $a \equiv 1 \pmod{3}$, cancellation of the h -terms leaves at most two members in each residue class modulo h . Both are impossible for a nonempty union of triples. The support is therefore empty, forcing $a = 1$. Thus A_k has Frobenius orbit length $2h$, and the displayed resultant is its irreducible minimal polynomial.

This also verifies the defining field used in Proposition 7.2. Put $\beta = \zeta + 4$. The identity $A_k = N(\beta)$ puts A_k in $\mathbb{F}_2(\beta)$, so the degree just proved shows that $\mathbb{F}_{2^{2h}} \subseteq \mathbb{F}_2(\beta)$. On the other hand $\beta \notin \mathbb{F}_{2^{2h}}$: otherwise $4 = \beta + \zeta$ would belong to that field, contrary to $\mathbb{F}_{16} \cap \mathbb{F}_{2^{2h}} = \mathbb{F}_4$. Hence $\mathbb{F}_2(\beta) = \mathbb{F}_{2^{4h}}$.

Writing $a_0 = 2^{3^{k-1}}$, one has

$$\Phi_{2 \cdot 3^k}(2) = a_0^2 - a_0 + 1 \equiv 3 \pmod{9}.$$

After removing its single factor of three, the cyclotomic divisor criterion gives the asserted prime support and valuations.

Fix $\ell \mid \Psi_k$. Proposition 7.1 gives $m_\ell \geq 4$. Propositions 2.3 and 7.2 give

$$m_\ell = 4 \iff A_k \notin (\mathbb{F}_{2^{2h}}^\times)^\ell.$$

Raising to $q-1$ is invertible on the current ℓ -Kummer quotient, because $q-1 \equiv -2 \pmod{\ell}$. Hence this is equivalent to the full ℓ -primary part appearing in $\text{ord}(M_k)$. Imposing it for every current prime gives assertion 2. Finally, μ_ℓ lies in $\mathbb{F}_{2^{2h}}$, so Kummer theory and Capelli's lemma identify non- ℓ -power of A_k with irreducibility of $R_k(X^\ell)$. $\square$

Proposition 7.4 (effective bound on the exceptional column). *Let $k \geq 2$, let $h = 3^k$, and let $\ell \mid \Psi_k$ be prime. Then*

$$m_\ell < 16h^4.$$

More precisely, let t be the least power of two satisfying

$$t \geq \lceil \log_2(h-1) \rceil.$$

Then $m_\ell \leq 4^t - 1$.

Proof. Put

$$K = \mathbb{F}_{4^h} = \mathbb{F}_4(\zeta), \quad C_a = \mathbb{F}_{4^t}, \quad L_a = KC_a = \mathbb{F}_{4^{ht}},$$

where $t = 2^{a-1}$. Choose a character $\chi : K^\times \rightarrow \mu_\ell$ of exact order ℓ , and define

$$\chi_a = \chi \circ N_{L_a/K}, \quad S_{a,k,\ell} = \sum_{c \in C_a} \chi_a(\zeta + c).$$

Because h is odd and t is a power of two, the fields K and C_a are linearly disjoint over $\mathbb{F}_4$, and $[L_a : K] = t$. This degree is prime to ℓ , so norm induces an isomorphism on ℓ -Kummer quotients. In particular, χ_a has kernel $(L_a^\times)^\ell$. No term $\zeta + c$ vanishes, and equality in the complex triangle inequality gives

$$\zeta + C_a \subseteq (L_a^\times)^\ell \iff S_{a,k,\ell} = 4^t.$$

The character χ is the primitive even Dirichlet character modulo $X^h + \omega$ over $\mathbb{F}_4[X]$. Write its L -polynomial as

$$L_\chi(T) = (1-T) \prod_{j=1}^{h-2} (1 - \alpha_j T).$$

The function-field Riemann hypothesis gives $|\alpha_j| = 2$, and constant-field extension raises each reciprocal root to its t -th power [13, Chapters 4, 8, and 9]. Comparing the degree-one coefficient after extension gives

$$S_{a,k,\ell} = - \left(1 + \sum_{j=1}^{h-2} \alpha_j^t \right), \quad |S_{a,k,\ell}| \leq 1 + (h-2)2^t.$$

If $2^t \geq h-1$, then

$$1 + (h-2)2^t < 4^t.$$

Thus some $c \in C_a$ makes $\zeta + c$ a non- ℓ -th power. Kummer-quotient injectivity from its defining field to L_a shows that the same is true in the defining field. The elements of C_a are exactly the finite numbers below 4^t , so $m_\ell \leq 4^t - 1$.

Finally, the least power of two t above $\lceil \log_2(h-1) \rceil$ satisfies

$$t < 2(\log_2 h + 1),$$

and hence $4^t < 16h^4$. This proves the stated bound. $\square$

The open statement D is therefore the family

$$D'_k : \quad \Psi_k \mid \text{ord}(M_k).$$

It is a current-level assertion, not a cumulative divisibility statement. The factor A_k in Proposition 7.2 is also essential: replacing it by a scalar product of separate translates gives the wrong norm.

The exceptional target fits the same cubical primitive-support coordinate as the multicomponent zero arm. Put $H = 2h$, $s = h/3$, and define on $\mathbb{F}_{2^H}^\times$

$$\sigma_2(x) = x^{2^h}, \quad \sigma_3(x) = x^{2^{2s}}, \quad \Delta_H(x) = \frac{x \sigma_2 \sigma_3(x)}{\sigma_2(x) \sigma_3(x)}.$$

Since $M_k = \sigma_2(A_k)/A_k$ and the two Frobenius maps commute,

$$\Delta_H(A_k) = \frac{\sigma_3(M_k)}{M_k} = M_k^{2^{2s}-1}.$$

For every $\ell \mid \Psi_k$, one has $\text{ord}_\ell(2) = 2h$, so $\ell \nmid 2^{2s} - 1$. Thus

$$\Psi_k \mid \text{ord}(M_k) \iff \Psi_k \mid \text{ord}(\Delta_H(A_k)).$$

Proposition 4.3 gives $|\text{im } \Delta_H| = \Phi_H(2) = 3\Psi_k$. This places D canonically in the primitive-support quotient, but does not prove generation: the displayed exponent is an invertible change of the same selected Kummer coordinate.

There is also a global cyclotomic-unit lift of the literal translate by four. It proves that the Kummer extension itself is nontrivial, while sharply separating that fact from the value of Frobenius at the selected prime.

Proposition 7.5 (marked exceptional cyclotomic unit). *Let $k \geq 2$, $h = 3^k$, and $m = 5 \cdot 3^{k+1}$. In $F = \mathbb{Q}(\mu_m)$, choose a primitive m -th root g , and put*

$$\zeta = g^5, \quad \theta = g^h, \quad \theta^5 = \zeta^h,$$

and let $\mathfrak{p} \mid 2$ be the prime selected by the reductions $\zeta \mapsto \zeta_k$ and $\theta \mapsto 4$. Put

$$\rho = \zeta/\theta, \quad u = 1 + \rho.$$

Then ρ has exact order m , and for every $\ell \mid \Psi_k$, and every extension $\mathfrak{P}$ of $\mathfrak{p}$ to $F(\mu_\ell)$,

$$m_\ell = 4 \iff \left(\frac{u}{\mathfrak{P}} \right)_\ell \neq 1.$$

Moreover,

$$u \notin (\mathbb{Q}^{\text{ab}, \times})^\ell.$$

Thus the global Kummer extension $F(\mu_\ell, u^{1/\ell})/F(\mu_\ell)$ is nontrivial; the remaining assertion is that its Frobenius at the Conway-selected prime is nontrivial.

Proof. The roots ζ and θ have orders 3^{k+1} and 15, respectively, and $\rho = g^{5-h}$ is primitive because $5-h$ is coprime to both three and five. The selected reduction is compatible: $\zeta_k^h = 4^5 = \omega$; if

$5a + hb = 1$, then $t = \zeta_k^a 4^b$ satisfies $t^5 = \zeta_k$ and $t^h = 4$, so evaluation $g \mapsto t$ defines the required prime. That prime $\mathfrak{p}$ has residue degree

$$\text{ord}_m(2) = \text{lcm}(2h, 4) = 4h.$$

Since $\varphi(m) = 8h$, there are exactly two primes over two; the compatible reductions select one of them. Also $\text{ord}_\ell(2) = 2h$, so the residue field already contains μ_ℓ .

The reduction of

$$\zeta + \theta = \theta(1 + \rho) = \theta u$$

is the literal element $\zeta_k + 4$. Because ℓ is coprime to 15, the reduction of θ is an ℓ -th power in the residue field. Hence u and the literal translate have the same selected ℓ -Kummer class. The power-residue formula and Propositions 7.2–7.3 prove the displayed equivalence.

The element u is a cyclotomic unit: $-\rho$ has conductor $2m$, which is not a prime power, and therefore

$$N_{F/\mathbb{Q}}(1 + \rho) = \Phi_m(-1) = \Phi_{2m}(1) = 1.$$

Its logarithmic height satisfies $h(u) \leq \log 2$. It is not a root of unity, since an embedding with $\rho = e^{2\pi i/m}$ has $|1 + \rho| > 1$. If $u = v^\ell$ for $v \in \mathbb{Q}^{\text{ab}, \times}$, then v is not torsion. Since $\text{ord}_\ell(2) = 2h \geq 18$, one has $\ell \geq 19$, and

$$h(v) = \frac{h(u)}{\ell} \leq \frac{\log 2}{19} < \frac{\log 5}{12}.$$

The last strict inequality is $2^{12} < 5^{19}$, while the lower bound $h(v) \geq (\log 5)/12$ is the theorem of Amoroso and Dvornicich [2]. This contradiction proves the global nonpower assertion. $\square$

The last statement is not yet the selected nonvanishing. Indeed, let $G = (\mathbb{Z}/m\mathbb{Z})^\times$ and let $J = \langle 2 \rangle$. Then $|J| = 4h$, $-1 \notin J$, and $G = J \sqcup (-J)$. If λ is the additive ℓ -Kummer logarithm after reduction at $\mathfrak{p}$, then characteristic two and $1 + \rho^{-a} = \rho^{-a}(1 + \rho^a)$ give

$$\lambda(1 + \rho^{2a}) = 2\lambda(1 + \rho^a), \quad \lambda(1 + \rho^{-a}) = \lambda(1 + \rho^a).$$

Here the inverse identity uses that the root-of-unity factor ρ^{-a} , whose order is coprime to ℓ , has trivial ℓ -Kummer class. Consequently every conjugate value is $2^j c$ for the single scalar $c = \lambda(u)$. All homogeneous circular-unit norm and orbit identities are compatible with $c = 0$, even though the marked global unit is not an ℓ -th power. Arm D is exactly the missing assertion $c \neq 0$.

The entire current circular-unit subgroup can nevertheless be identified. This removes a possible third obstruction and leaves two exact ray-class factors.

Proposition 7.6 (exact exceptional circular-ray obstruction). *Put $n = 3^{k+1} = 3h$, $f = 5n$, and $E = \mathbb{Q}(\mu_f)$. Choose roots z, λ of orders n, five , put $\omega = z^h$, and let*

$$L = \mathbb{Q}(z, \lambda + \lambda^{-1}).$$

Select $\mathfrak{P} \mid 2$ by $z \mapsto \zeta_k$ and $\lambda + \lambda^{-1} \mapsto \omega$. Let $k_{\mathfrak{P}}$ be the residue-field multiplicative group, and let E_L and C_L be the global and Sinnott circular units. For $\ell \mid \Psi_k$, write a subscript ℓ for the ℓ -primary projection after reduction. Then

$$\overline{C}_{L,\ell} = \langle A_{k,\ell} \rangle = \langle M_{k,\ell} \rangle.$$

Consequently

$$[k_{\mathfrak{P},\ell} : \langle M_{k,\ell} \rangle] = |\ker(\text{Cl}_L(\mathfrak{P})_\ell \longrightarrow \text{Cl}_L)_\ell| \cdot |(\overline{E}_L/\overline{C}_L)_\ell|. \quad (**)$$

The second factor divides the ℓ -part of the class number of the maximal real subfield L^+ . Hence D at (k, ℓ) is equivalent to simultaneous triviality of the principal $\mathfrak{P}$ -ray ℓ -kernel and the reduced unit/circular-unit ℓ -quotient.

Proof. The field L has degree $4h$. Frobenius at two has order $2h$, so there are exactly two primes over two and $k_{\mathfrak{P}} \simeq \mathbb{F}_{2^{2h}}^\times$. Put $x = \omega z$ and

$$U = \omega N_{E/L}(1 - x\lambda) = \omega(1 - x\lambda)(1 - x\lambda^{-1}).$$

The root $x\lambda$ has exact conductor f , so $U \in C_L$, and direct reduction gives

$$\overline{U} = \omega(1 + \omega\zeta_k\omega + \omega^2\zeta_k^2) = \zeta_k^2 + \zeta_k + \omega = A_k.$$

Let $\pi_\ell : k_{\mathfrak{P}} \rightarrow k_{\mathfrak{P},\ell}$ be the canonical projection to the full ℓ -Sylow subgroup. Use the standard normed circular-unit generators from conductors $d \mid f$. If the 3-part of d is at most 3^k , the residue degree in $L \cap \mathbb{Q}(\mu_d)$ is at most $2h/3$. Because $\text{ord}_\ell(2) = 2h$, its residue group has no ℓ -part, so π_ℓ kills every such generator. This includes the pure conductor-five terms.

At full 3-conductor $d = n$ without the factor five, every unit $e \in \mathbb{Q}(\mu_n)$ satisfies $\bar{e}/e \in \mu_\infty$: all conjugates of this quotient have absolute value one, so Kronecker's theorem applies. The 2^h -Frobenius is complex conjugation on this field. The root-of-unity factor has order prime to ℓ , and hence

$$\pi_\ell(e)^{2^h} = \pi_\ell(\bar{e}) = \pi_\ell(e).$$

Since $\ell \nmid 2^h - 1$, exponentiation by $2^h - 1$ is an automorphism of the entire ℓ -Sylow subgroup, forcing $\pi_\ell(e) = 1$. Only the top-conductor generators remain; they form two decomposition-group orbits. More precisely, the standard top-conductor normed generators are, up to roots of unity, the $\text{Gal}(L/\mathbb{Q})$ -conjugates of U .

Let r fix z and induce the nontrivial automorphism on $\mathbb{Q}(\lambda + \lambda^{-1})$. The two orbits are coupled by

$$U r(U) = \omega^2 \prod_{a=1}^4 (1 - x\lambda^a) = \omega^2 \Phi_5(x) \in \mathbb{Q}(\mu_n)^\times.$$

The right side has trivial full ℓ -primary projection by the preceding full-3-conductor argument. Hence $\pi_\ell(r(U)) = \pi_\ell(U)^{-1}$. Within each decomposition orbit, Frobenius sends $\pi_\ell(U)$ to its successive 2^j -th powers, and the second orbit gives their inverses. Thus every top-conductor circular unit projects into the cyclic subgroup generated by $\pi_\ell(\overline{U}) = A_{k,\ell}$. Since U itself is circular, this proves the first equality at the full primary valuation.

Since $M_k = A_k^{2^h-1}$ and $\ell \nmid 2^h - 1$, the A_k - and M_k -subgroups agree. Factoring the two subgroup indices gives (**). The first factor is the principal ray kernel. For odd ℓ , possible Hasse-unit and circular-unit normalization factors in Sinnott's formula are powers of two, so the ℓ -part of E_L/C_L equals the ℓ -part of the class number of L^+ ; reduction can only decrease it. This proves the divisibility and the final equivalence; see [15]. $\square$

The local coordinates in the cubic and exceptional arms are exactly orthogonal, so their ray obstructions cannot cancel or force one another.

Proposition 7.7 (current C/D orthogonality). *Let $h = 3^k$, $q_0 = 2^{h/3}$, and $q = q_0^3 = 2^h$. Put*

$$U_C = \ker \left(N_{\mathbb{F}_q/\mathbb{F}_{q_0}} \right), \quad U_D = \ker \left(N_{\mathbb{F}_{q^2}/\mathbb{F}_q} \right).$$

Then

$$|U_C| = q_0^2 + q_0 + 1, \quad |U_D| = q + 1, \quad \gcd(|U_C|, |U_D|) = 1.$$

Moreover,

$$\mathbb{F}_{q^2}^\times = \mathbb{F}_q^\times \times U_D, \quad N_{\mathbb{F}_{q^2}/\mathbb{F}_q}(b, u) = b^2,$$

and the cubical operator Δ_{2h} is trivial on $\mathbb{F}_q^\times$ while its image lies in U_D . Thus every current C -primary class lies in the factor killed by Δ_{2h} , whereas every current D -primary class is killed by the local norm to the cubic field.

Proof. One has

$$q + 1 = (q_0 + 1)(q_0^2 - q_0 + 1).$$

The number $q_0^2 + q_0 + 1$ is coprime to $q_0 + 1$, by reduction at $q_0 = -1$, and coprime to $q_0^2 - q_0 + 1$, because their difference is $2q_0$ and both are odd. This proves the coprimality. Since $\gcd(q-1, q+1) = 1$, the two displayed subgroups give the direct-product decomposition. The norm is squaring on $\mathbb{F}_q^\times$ and is one on U_D . Finally, the definition of Δ_{2h} gives one on the subfield $\mathbb{F}_q^\times$, while the identity $\Delta_{2h}(x) = M(x)^{2^{h/3}-1}$ places its image in U_D . $\square$

In particular, the shared prime over two supplies no local norm implication between the two principal-ray factors. A proof in either arm must evaluate the actual image of global units in its own coprime local factor.

The exceptional factors have a further faithful-character localization. In the maximal real field $K = \mathbb{Q}(\mu_{3^{k+1}})^+ \mathbb{Q}(\sqrt{5})$, the prime over two is totally inert and $G = \text{Gal}(K/\mathbb{Q}) = \langle \sigma \rangle$ has order $2h$, with $\sigma = \text{Frob}_2$. On the first current ℓ -layer, both factors in $(**)$ lie in the faithful character

$$\chi_\ell(\sigma) = 2 \pmod{\ell}.$$

For the cubic step, $\tau = \sigma^{2h/3}$ acts on this line by a primitive cube root r , so

$$V^{(\tau)} = 0, \quad (1 + \tau + \tau^2)V = (1 + r + r^2)V = 0$$

even when $V \neq 0$. Ambiguous-class, genus, and norm formulas therefore annihilate precisely the faithful component they would need to control. Fixed-prime Iwasawa stabilization does not supply a uniform substitute, because ℓ and its faithful character are born anew at each level.

Proposition 7.8 (prime current factor). *Let $k \geq 2$, put*

$$s = 3^{k-1}, \quad a = 2^s, \quad \Lambda_k = a^2 - a + 1 = 3\Psi_k,$$

and define $P_k = M_k^{a+1}$. Then

$$P_k^{\Lambda_k} = 1, \quad [\mathbb{F}_2(P_k) : \mathbb{F}_2] = 2 \cdot 3^k,$$

and exponentiation by $a+1$ preserves every primary part supported on Ψ_k . Consequently D'_k holds whenever Ψ_k is prime. In particular, $\Psi_2 = 19$ and $\Psi_3 = 87211$ are prime, so D'_2 and D'_3 hold.

Proof. Write $h = 3^k$ and $q = a^3 = 2^h$. Since

$$q + 1 = (a + 1)(a^2 - a + 1),$$

the norm-one relation $M_k^{q+1} = 1$ gives $P_k^{\Lambda_k} = 1$. Moreover, $\gcd(\Lambda_k, a + 1) = 3$ and $v_3(\Lambda_k) = 1$, hence $\gcd(\Psi_k, a + 1) = 1$. This proves the assertion about primary parts.

The congruences

$$a \equiv h - 1, \quad a^2 \equiv h + 1, \quad a^3 \equiv -1, \quad a^4 \equiv 2h + 1 \pmod{3h}$$

give, by direct Frobenius expansion,

$$P_k = \frac{\zeta^4 + \zeta + \omega}{\omega^2 \zeta^4 + \zeta^3 + 1}.$$

Also $P_k^q = P_k^{-1}$. If $P_k \in \mathbb{F}_q$, then $P_k = 1$; cross-multiplication would give

$$\zeta^{h+4} + \zeta^h + \zeta^3 + \zeta + 1 = 0.$$

If $P_k \in \mathbb{F}_{a^2}$, its order divides $\gcd(\Lambda_k, a^2 - 1) = 3$, so $P_k \in \{1, \omega, \omega^2\}$. The two remaining values similarly give

$$\zeta^{h+3} + \zeta = 0$$

or

$$\zeta^{h+4} + \zeta^{h+3} + \zeta^4 + \zeta^3 + \zeta + 1 = 0.$$

Each displayed polynomial is nonzero and has degree below $2h$, contradicting the minimal polynomial $X^{2h} + X^h + 1$ of ζ .

Every proper subfield of $\mathbb{F}_{2^{2h}}$ lies in $\mathbb{F}_q$ or $\mathbb{F}_{a^2}$. Thus P_k has degree $2h$. If Ψ_k is prime, its order, a divisor of $3\Psi_k$, cannot divide three and must therefore contain Ψ_k . The primary-part assertion transfers this divisibility to $\text{ord}(M_k)$. $\square$

The cubic and exceptional current factors are coprime:

$$\gcd(\Phi_{3^k}(2), \Phi_{2 \cdot 3^k}(2)) = 1.$$

Thus arm C and arm D occupy distinct primary coordinates. Norm recursion in the cubic tower cannot by itself transfer the missing order from one arm to the other. Their common Conway ancestry is relevant only through the explicitly selected additive coefficients.

Proposition 7.9 (common toric cubic dynamics). *Over a field of characteristic two, for $z \neq 0$ put*

$$\varpi(z) = z + z^{-1}, \quad D(X) = X^3 + X, \quad F_c(U) = U^3 + cU^2 + cU,$$

where $c^2 + c = 1$. Then

$$D(\varpi(z)) = \varpi(z^3), \quad F_c(U) = D(U + c) + c^2,$$

and

$$F_{c^2}(F_c(U)) = D(D(U + c)) + c.$$

In the literal cubic tower choose alternating $c_j \in \mathbb{F}_4$ with $c_j^2 + c_j = 1$ and $c_{j-1} = c_j^2$, and put $u_j = \gamma_j + c_j$. Then

$$F_{c_j}(u_j) = u_{j-1}.$$

Consequently the selected C ancestry is the inversion quotient of the 3-division tree on $\mathbb{G}_m$, while this auxiliary recursion used in the analysis of D is a two-periodic affine coordinate on the same toric tree. This does not identify u_k with the exceptional target A_k , whose Kummer class remains the one in Theorem 7.3.

Proof. Expanding $(z + z^{-1})^3 + (z + z^{-1})$ in characteristic two leaves $z^3 + z^{-3}$. The second identity is another direct expansion using $c^2 + c = 1$, which also gives $c^3 = 1$. Applying it first with c and then with c^2 , and using $c^4 = c$, gives the third identity. Finally, $\gamma_j^3 + \gamma_j = \gamma_{j-1}$, so the second identity gives

$$F_{c_j}(u_j) = D(\gamma_j) + c_j^2 = \gamma_{j-1} + c_{j-1} = u_{j-1}.$$

$\square$

This exact identification rules out direct identification with several tempting division systems. The map D is not additive:

$$D(X + Y) - D(X) - D(Y) = XY(X + Y),$$

and its degree is three, whereas a nonconstant Drinfeld or Carlitz endomorphism in characteristic two is an additive polynomial of power-of-two degree. Nor is this a smooth elliptic Lattès system: infinity is a totally invariant exceptional point of the polynomial D , while a finite quotient of a separable elliptic isogeny of degree greater than one has no exceptional point, since the inverse images upstairs grow with every iterate and quotient fibres stay bounded. The relevant algebraic group is $\mathbb{G}_m$, the smooth group of a nodal cubic, not a hidden smooth elliptic division tower.

The natural characteristic-zero Dickson lift $X^3 - 3X$ has the same semiconjugacy with cubing. The selected C -preimages come from roots of unity and therefore are preperiodic and have canonical height zero. The forward-orbit primitive-divisor theorem of Ingram and Silverman assumes an infinite orbit and therefore does not apply [7]. More fundamentally, these viewpoints still end at the same selected values $\Omega_{k,\ell}$; none evaluates them, and the exceptional target remains separate as stated above.

8 Boundedness and the remaining theorem

Lenstra asked whether the finite excesses are absolutely bounded [8]. Conjecture 1.1 would give the bound four, but boundedness itself is weaker than any maximal-order assertion above.

For $a \geq 0$, let

$$C_a = \mathbb{F}_{2^{2^a}}$$

be the complete finite Conway subfield at that level. For an odd prime p , put $x_p = \kappa_{f(p)}$ and $L_{p,a} = \mathbb{F}_2(x_p)C_a$.

Proposition 8.1 (finite-line criterion for absolute boundedness). *For every p and a ,*

$$m_p < 2^{2^a} \iff \text{there exists } c \in C_a \text{ such that } x_p + c \notin L_{p,a}^p,$$

Consequently,

$$\sup_p m_p < \infty \iff \text{for some fixed } a, \text{ every } p \text{ has such a } c \in C_a.$$

Here $L^p = \{y^p : y \in L\}$ for a field L .

Proof. For $c \in C_a$, let $F_c = \mathbb{F}_2(x_p + c)$. Since $x_p = (x_p + c) + c$, one has $F_c C_a = L_{p,a}$. The degree $[L_{p,a} : F_c]$ divides $[C_a : \mathbb{F}_2] = 2^a$, so it is prime to the odd prime p . Extension to $L_{p,a}$ is therefore injective on p -Kummer quotients: restriction followed by norm is multiplication by this invertible degree. Thus $x_p + c$ is a p -th power in its defining field exactly when it is a p -th power in $L_{p,a}$.

The finite numbers below 2^{2^a} are exactly C_a , which proves the first equivalence from the definition of m_p . A fixed a gives the uniform bound $2^{2^a} - 1$. Conversely, choose a with 2^{2^a} larger than a given uniform bound. $\square$

The proposition says that boundedness is uniform nonsaturation of one fixed finite affine line. It does not require any of the particular translates $0, 1, 4$ to have maximal primary order. No fixed line is presently known to work for all primes.

The universal frontier can now be stated without auxiliary machinery:

1. prove generation by the synchronized structural norm T_h , including the Conway–Fermat equalities $\delta_n = F_n$;
2. prove full primary order for the marked projective unit $y_{r,a}$ on every ordinary odd spine;
3. prove that every selected Gaussian period γ_k , equivalently every selected norm-one increment η_k , is primitive;
4. prove $\Psi_k \mid \text{ord}(M_k)$ for every exceptional level.

Each item is a one-dimensional selected Kummer nonvanishing problem after the reductions above. Generic degree, trace, norm, or unmarked reciprocity data either determine a different coordinate or reconstruct the same selected coordinate. They do not supply the missing nonvanishing.

9 Formal and computational verification

The mathematical statements and the evidence are deliberately separated.

9.1 Formal boundary

The Lean 4 and Mathlib development [5, 14] proves the abstract cyclic-group power criterion, its full-primary order form, quotient-order factorization, the corrected norm identity, the exceptional lower-bound interface, and a collection of polynomial and finite arithmetic lemmas used by the reductions. It also proves the torus/Dickson semiconjugacy, the alternating two-step cubic conjugacy, and the order-four coordinate symmetry of the supersingular Conway curve. The two-face, three-face, and two-prime cubical identities support the common primitive-support coordinate used in the zero and exceptional arms. It checks the recorded factorizations of Ψ_k for $2 \leq k \leq 6$, the primality of the modest factors through $k = 4$, and exact order-of-two screens for the recorded factors.

The identifier `DPrimeTarget` defines

$$\forall k \geq 1, \quad \Psi_k \mid \text{ord}(M_k).$$

It is not assumed and is not proved. The formal files verify ingredients of Theorem 3.1; they do not contain an end-to-end proof of Conjecture 1.1.

The formal project is built with

```
cd formal
lake build Ogdoad.Excess
```

The relevant source is `formal/Ogdoad/Excess.lean`.

9.2 Finite evidence

The finite evidence has the following exact status.

- The 126 values for odd primes $3 \leq p \leq 709$ are imported from OEIS A380496 and checked against the complete vendored sequence [11]. They are external tabulated data, not a proof of a universal pattern.

- The values $m_{359} = 1$, $m_{719} = 1$, and $m_{727} = 1$ have local exact finite-field certificates. The first two use marked Hilbert and crossed-norm payloads; the last uses a self-contained degree-2420 polynomial certificate. The next unsupported implementation row, $p = 733$, is an operational guardrail rather than a mathematical boundary.
- The cubic assertions C_1, C_2, C_3 are proved by exact order arguments. Levels C_4, C_5 have complete finite certificates; C_6 retains a source-assisted factorization boundary, and later checked factors are evidence only.
- The exceptional levels D'_2, D'_3 follow analytically from Proposition 7.8. Levels D'_4, D'_5 have complete finite certificates. Level D'_6 retains a source-assisted primality boundary; later partial factorizations do not certify a level.

The maintained finite-field entry points are

```
python3 experiments/ordinal_excess_probe.py
python3 experiments/cyclotomic_3k_family.py
python3 experiments/exception_column_m4.py
python3 experiments/ordinary_359_hilbert_certificate.py --full
python3 experiments/ordinary_719_crossed_certificate.py --full
python3 experiments/ordinary_727_certificate.py
```

The full 359 and 719 certificates are intentionally expensive. A successful finite run proves only its named row or level.

10 Conclusion

The finite excess problem is not presently a problem of discovering another candidate sequence. Its exact candidate is Conjecture 1.1, and Theorem 3.1 reduces it to four explicit selected-order assertions. The reductions explain both the strength of the numerical pattern and the persistence of the proof gap: all coarse finite-field invariants are compatible with more than one value of the selected Kummer coordinate.

The most economical remaining target is therefore direct. One must prove nonvanishing of the recursively selected Euler quotient in each of the four arms, with full primary valuation. Until those four statements are proved, the universal $0/1/4$ rule and even absolute boundedness remain open.

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